

Quantum Division Circuit Based on Restoring Division Algorithm

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Abstract— This paper presents a new quantum circuit for implementation of division operation in n-qubit quantum processors based on restoring division algorithm. In the proposed circuit we tried to reduce the number of ancilla qubits as many as possible using quantum fourier transform so that the circuit could be easily generalized for the case of any arbitrary values of dividend and divisor.

Keywords—quantum division; restoring division; quantum fourier transform; quantum addition

I. INTRODUCTION

Fundamental algorithms in the quantum computation field such as Shor's [1], Grover's [2] and Quantum fourier transform [3] proved that quantum algorithms can be much more efficient than their classical counterparts in solving certain problems. These algorithms gave enough motivation to scientists for establishing the idea of a quantum computer which exploits the extraordinary features of quantum world such as entanglement and superposition to speed up calculations.

Different elementary quantum networks for arithmetic operations such as addition, subtraction, multiplication, shift operation, bit string comparison, etc have been proposed in recent years [4], [5], [6], [7], [8], [9].

Neither of the papers mentioned above have suggested an explicit quantum circuit for division operation, except for [8] where it is suggested that a quantum divider could be implemented using the algorithm shown in Fig.1 exploiting the basic arithmetic submodules such as quantum comparator, quantum adder and quantum subtractor. However implementing a quantum divider circuit based on this algorithm is prone to the following problems:

- 1) The number of circuit components (here unitary quantum operators) in a quantum circuit based on an iterative algorithm depends linearly on the number of iterations and for the mentioned algorithm the number of iterations depends on the magnitude of input values. In general for divideng two n-qubit numbers using this algorithm the number of iterations would be (N/D) where it is something between 0 and 2^n-1 , hence the average number of iterations for n qubit division is $1/2 \times (2^n-1)$.
- 2) Implementation of each iteration of this algorithm needs one comparison, one subtraction and one addition. The optimum quantum bit string comparator for comparing two n-qubit numbers a and b is proposed in [10] where it is shown that for this operation $2n$ ancilla bits are necessary. Moreover in each step of the comparison some of the ancilla bits change which means that an extra circuitry should be used to reversibly set back this qubits to zero.

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Q=0
while N>= D do
    N=N-D
    Q=Q+1
end
* N = numerator, D = denominator, Q = quotient .

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Figure 1. The pseudo code for division operation used in [8].

- 3) This algorithm only works for division of integer numbers.

In this paper we present a quantum circuit for division operation of two numbers N (numerator) and D (denominator) prepared in n-qubit fixed-point binary format using the restoring division algorithm with the intellectual choice of rotation based addition proposed by [5].

II. THEORETICAL BACKGROUND

In this section we intend to take a brief look at some of the basic quantum circuits such as quantum shift register, quantum adder, quantum subtractor and quantum fourier transform which will be useful in development of our quantum division circuit.

A. Quantum Shift Register

One of the building blocks that we will frequently use in our quantum divider is a quantum shift register. Suppose that we have stored a number in an n-qubit register $|b\rangle = |b_n \dots b_2 b_1\rangle$, where each of the qubits is either $|0\rangle$ or $|1\rangle$ and we want to shift the register k times to the left. Due to the irreversibility of the classical shift operation, the information stored in the highest k qubits are lost when the register is left shifted k times. In order to make the shift operation reversible we use k ancilla qubits. The quantum circuit proposed in [9] for left shift operation is shown in Fig.2.

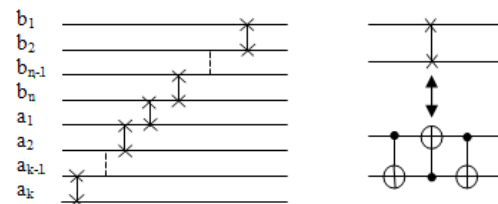


Figure 2. The quantum shift left register composed of swap gates. Register $|b\rangle = |b_n \dots b_2 b_1\rangle$ is shifted using k ancilla qubits. Note that the ancilla qubits must be initially set to zero.

Each swap operation transforms the state of the register as $|b\rangle = |a_k a_{k-1} \dots a_1; b_n \dots b_2 b_1\rangle \rightarrow |b\rangle = |a_{k-1} a_k \dots a_1; b_n \dots b_2 b_1\rangle \dots \rightarrow |b\rangle = |a_{k-1} a_{k-2} \dots b_n; b_{n-1} \dots b_1 a_k\rangle$.

B. Quantum Fourier Transform

Quantum fourier transform (QFT) blocks are used in carry save quantum addition operation. Here we present a quick review of QFT [11] which will be exploited to fulfill quantum division.

Suppose we have prepared an n-qubit quantum state $|b\rangle = |b_n \dots b_2 b_1\rangle$, where $|b_i\rangle$ s are either $|0\rangle$ or $|1\rangle$. Then the QFT of $|b\rangle$ is the mapping :

$$|b\rangle \rightarrow |\phi_n(b)\rangle \otimes \dots \otimes |\phi_2(b)\rangle \otimes |\phi_1(b)\rangle \quad (1)$$

Where :

$$|\phi_k(b)\rangle = (|0\rangle + e(b/2^k) |1\rangle) \quad (2)$$

and

$$e(x) = e^{2\pi i x} \quad (3)$$

Also note that :

$$e(b/2^k) = e(0.b_k \dots b_1) \quad (4)$$

A circuit for computing QFT is depicted in Fig.3. The circuit is composed of conditional rotation and Hadamard gates.

The Hadamard gate represented by H, operates on a single qubit and transforms the state $\alpha|0\rangle + \beta|1\rangle$ to $((\alpha + \beta)|0\rangle + (\alpha - \beta)|1\rangle)$.

$$H \equiv \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (5)$$

Conditional k rotation gate represented by R_k , performs the phase rotation on two qubits and transforms the state $\alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$ to $\alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + e(1/2^k)\delta|11\rangle$.

$$R_k \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e\left(\frac{1}{2^k}\right) \end{pmatrix} \quad (6)$$

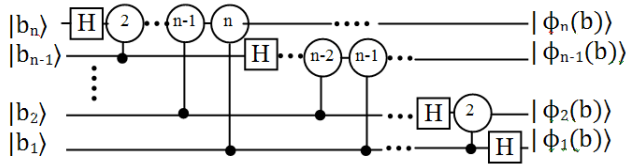


Figure 3. Quantum fourier transform circuit .

To see what happens in the quantum fourier transform circuit, we will follow a single qubit like $|b_n\rangle$:

$$\begin{aligned} |b_n\rangle &\rightarrow (|0\rangle + e(0.b_n) |1\rangle) && \text{Hadamard gate} \\ &\rightarrow (|0\rangle + e(0.b_n b_{n-1}) |1\rangle) && R_2 \text{ rotation} \\ &\vdots \\ &\rightarrow (|0\rangle + e(0.b_n b_{n-1} \dots b_1) |1\rangle) && R_n \text{ rotation} \\ &= \phi_n(b) \end{aligned}$$

C. Quantum addition and subtraction

Here we want to explain a carry save quantum addition operation without the use of extra qubits which was proposed by [5]. We will benefit from this circuit in our quantum division circuit.

The quantum addition circuit is very similar to QFT as it uses conditional rotations in its structure. Suppose we have two n-qubit numbers $|a\rangle$ and $|b\rangle$. The circuit in Fig.3 takes $|b\rangle$ and the QFT of $|a\rangle$ namely $|\phi(a)\rangle$ as its inputs, and gives $|b\rangle$ and the QFT of addition ($|\phi(a+b)\rangle$) in its output. By applying inverse QFT to $|\phi(a+b)\rangle$ we can get $|a+b\rangle$ which is the expected result for addition of these two numbers.

In order to understand the functionality of this circuit we will follow what each gate does to the qubit $|\phi_n(a)\rangle$.

$$\begin{aligned} |\phi_n(a)\rangle &\rightarrow (|0\rangle + e(0.a_n \dots a_1 + 0.b_n) |1\rangle) && R_1 \text{ rotation} \\ &\rightarrow (|0\rangle + e(0.a_n \dots a_1 + 0.b_n b_{n-1}) |1\rangle) && R_2 \text{ rotation} \\ &\vdots \\ &\rightarrow (|0\rangle + e(0.a_n \dots a_1 + 0.b_n b_{n-1} \dots b_1) |1\rangle) && R_n \text{ rotation} \\ &= |\phi_n(a+b)\rangle \end{aligned}$$

It is clear that the subtraction operation could be implemented by a little modification in R_k . If we replace the element $e(1/2^k)$ in R_k matrix with $e(-1/2^k)$ then the quantum adder circuit would act as a quantum subtractor whose output will become $|b\rangle \otimes |\phi(a-b)\rangle$.

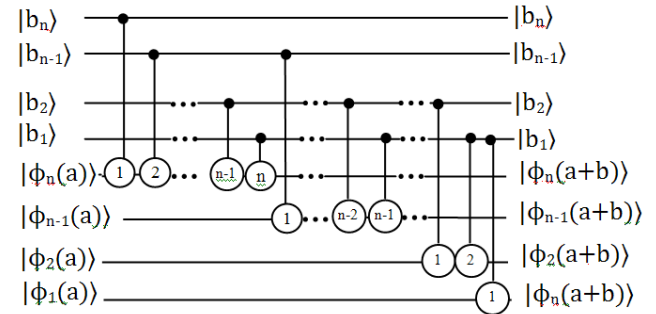


Figure 4. Quantum carry save adder.

III. QUANTUM DIVISION

In this section, a quantum circuit for division will be explicitly derived. The division operation is implemented using the restoring division algorithm. In Fig.5 you see a pseudo code for this algorithm. This algorithm is used to divide two fixed- point fractional numbers coded in n qubits.

We have chosen restoring division because it has an straightforward implementation in terms of quantum gates considered in previous section, exploiting only n extra qubits to implement the division operation reversibly.

Suppose we want to divide N(numerator) by D(denominator) and restore the division result in registers Q(quotient) and P(partial remainder). According to the algorithm registers P and D must have twice the word length of N and Q. At first the lowest n-qubits of P have been prepared equal to N and its highest n-qubits are set to zero. Also register D is initiated such that its highest n-qubits contain the denominator's value and its lowest n-qubits are set to zero. First of all P is multiplied by 2 and then decremented by D. If the result is positive then the qubit $q(n)$ in the quotient is set to 1, otherwise it is 0 and D should be added to P. This procedure is repeated n times and each time one of the qubits of quotient is determined. Eventually the division results, quotient and remainder will be available respectively in Q and highest n-qubits of P.

An example will clarify this procedure. Suppose $A=7=(111)_2$ and $B=2=(010)_2$ and we want to calculate quotient and remainder of operation A/B .

Initialization :

$$Q = (000)_2$$

$$P = (000111)_2$$

$$D = (010000)_2$$

First iteration:

$$P = 2P - D = (111110)_2 = \text{negative}$$

$$Q = (000)_2$$

$$P = P - D = (001110)_2$$

Second iteration:

$$P = 2P - D = (001100)_2 = \text{positive}$$

$$Q = (010)_2$$

Third iteration:

$$P = 2P - D = (001000)_2 = \text{positive}$$

$$Q = (011)_2$$

Result:

$$\text{Remainder} = \text{upper half of } P = (001)_2 = 1$$

$$\text{Quotient} = Q = (011)_2 = 3$$

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P = N
D = D << n      * P and D need twice the word width of N & Q
For i = n . . . 1 do * for example 31 . . . 0 for 32 bits
    P = 2P - D    * trial subtraction from shifted value
    If P >= 0 then
        q(n) = 1
    else
        q(n) = 0
        P = P + D
    end
end

* N= Numerator , D = Denominator, n= # bits ,
* P = Partial remainder, q(i) = bit # i of quotient
* N>D

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Figure 5. Pseudo code for restoring division algorithm.

Fig.6 illustrates the quantum divider based on restoring division algorithm. The input registers are $|P\rangle = |p_{2n} \dots p_1\rangle$, $|D\rangle = |d_n \dots d_1\rangle$, $|Q\rangle = |q_n \dots q_1\rangle$. Initially lowest n qubits of $|P\rangle$ contain the value of numerator and highest n qubits are zero (dedicated for n shift operations), $|Q\rangle$ is zero at first and $|D\rangle$ contains the value of denominator. In Fig.5 it was assumed that $|D\rangle$ is a $2n$ -qubit register, but it is clear that all of the qubits in the lower half of $|D\rangle$ are zero so they have no influence in addition and subtraction operations in the algorithm, therefore in our quantum divider, $|D\rangle$ has only n qubits which contain the value of denominator.

Here we want to describe the evolution of qubits in Fig.6. The left shift(LSH) module serves as a multiplier of $|P\rangle$ by 2. In order to subtract $|D\rangle$ from the highest n qubits of $|2P\rangle$, first we take QFT from highest n qubits of $|2P\rangle$ and then we apply a set of conditional rotations conditioned on qubits of $|D\rangle$ as described in previous section and after that we take inverse QFT. Now it is time to check whether $|(2P-D)\rangle$ is positive or negative. If it is positive we must set the qubit $|q_n\rangle$ which was initially in the state $|0\rangle$ otherwise we have to leave it unchanged. This task is simply done using a CNOT gate. The control qubit in CNOT is the most significant qubit of $|P\rangle$ namely $|p_n\rangle$ and the target qubit is $|q_n\rangle$. If the $|(2P-D)\rangle$ was diagnosed to be negative it should be set back to the previous state $|2P\rangle$ by simply adding $|D\rangle$ to the highest n qubits of $|(2P-D)\rangle$. It is an easy job because $|q_n\rangle$ contains the inverse of the most significant qubit in $|(2P-D)\rangle$, hence the addition could be conditioned on $|q_n\rangle$.

This structure should be repeated n times to fulfill the division operation. Eventually $|Q\rangle$ will contain the value of quotient and highest n qubits of $|P\rangle$ will contain the remainder value.

Also note that we have used only n ancilla qubits just for left shift operation.

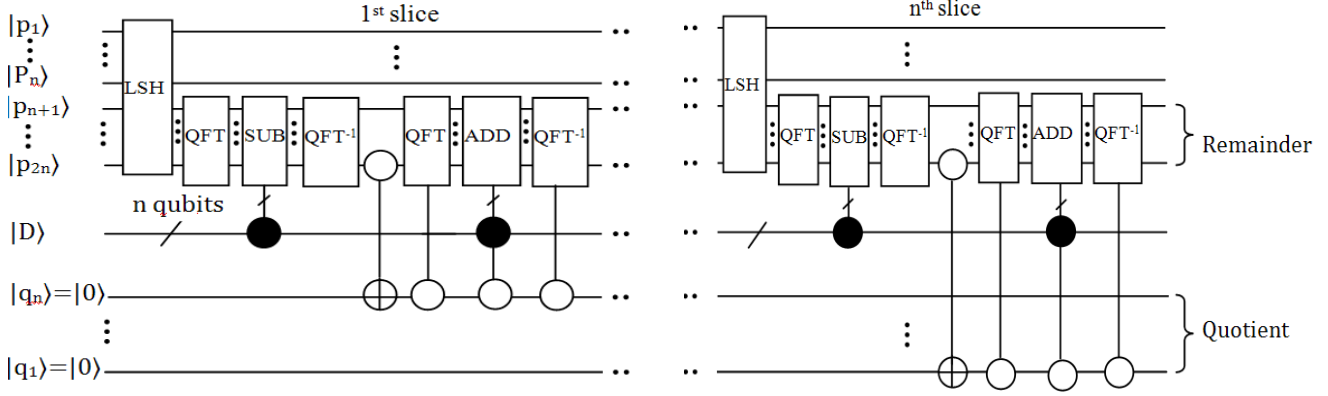


Figure 6. Quantum divider circuit based on restoring division algorithm.

At this stage it would be very useful to evaluate the cost of implementation of the quantum divider circuit. For n qubit division we have n shift operations, n additions, n subtractions, $2n$ quantum fourier transforms, $2n$ inverse quantum fourier transforms and n CNOT gates for controlling qubits of $q(i)$.

Shift operation involves $2n-1$ swap gates where each swap is made up of 3 CNOT operations. The total number of operations in shift registers would be equal to $n \times (2n-1) \times 3 \approx 6n^2$.

Each of the quantum addition, quantum subtraction, quantum fourier transform and inverse quantum fourier transform consist of $1/2n(n+1)$ operations where the total number of operations related to these four modules would be $(n + n + 2n + 2n) \times 1/2n(n+1) \approx 3n^3$ operations.

Therefore the total number of operations needed to implement quantum division is exactly $3n^3 + 6n^2 + n$.

Barneco, et. al. [12] proved that in presence of decoherence the approximate quantum fourier transform (AQFT) where only rotations R_k with $k \leq \log_2^n$ are used could be more accurate than a complete quantum fourier transform. This argument also implies to quantum addition and quantum subtraction because of the great similarities. By this assumptions the total number of operations would become $6n^2 \log_2^n + 6n^2 - 2n \approx 6n^2 \log_2^{2n}$.

Since the conditional rotation operations commute with each other, the quantum adder can be implemented by first applying all R_1 rotations and then R_2 rotations etc. Therefore in a quantum computer capable of implementing numerous independent gate operations simultaneously the run time of division will decrease proportionally [13].

IV. CUNCLUSION

Although there have been a thorough study in the field of quantum implementation of elementary arithmetic operation, no

one had addressed an explicit quantum circuit for division operation before. In this work we have proposed a quantum implementation of division operation for division of two n -qubit fixed point numbers in binary format based on restoring division algorithm, exploiting quantum shift register, quantum fourier transform and quantum addition. The division operation is fulfilled by employing just n ancilla qubits. The circuit also benefits from massive parallelism contained in the rotation based quantum addition and subtraction submodules [5].

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